

The 3-adic Shadow of a Collatz Orbit: Divergence as Sustained Compression

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Abstract

Let T be the Terras form of the Collatz map and let $j = j_T(n)$ count odd steps among the first T . The exact orbit identity $2^T T^T(n) = 3^j n + d$ has correction term d depending only on the parity word of n ; reducing modulo 3^j eliminates the n -term entirely, so *the 3-adic residue of an orbit is a function of its parity word alone*. We develop the consequences, all machine-verified in Lean 4: the correction obeys the cocycle law $d(w_1 w_2) = 3^{j_2} d(w_1) + 2^{|w_1|} d(w_2)$, exhibiting the integer dynamics as the $\mathbb{Z}_2 \times \mathbb{Z}_3$ skew product in elementary form; orbits whose values remain $\geq N$ survive without descent only at odd density $\geq \log 2 / \log \frac{3N+1}{N}$, so divergent orbits are pinned to the critical density $\log 2 / \log 3$, at which the 3-adic digit production rate $\delta \log_2 3$ equals exactly 1 bit per step; and the residues attainable by surviving windows of length T number at most $3^T / 2^{\lfloor 17(T-1)/27 \rfloor} \approx 2^{0.955T}$ inside fibers of more than 2^{T-1} classes — the precise 3-adic mirror, with identical constants, of Terras’ theorem in finite form. Together these give a quantitative reduction: a divergent Collatz orbit is exactly an integer that compresses its own 3-adic digit stream by $\approx 4.5\%$, at every scale, forever. Experiments sharpen the picture: the true shadow entropy lies below the binomial bound; the full-window shadow nearly determines the half-window shadow (a strong cross-scale rigidity predicted by the cocycle); and Sturmian parity words — forced on slowly divergent orbits — occupy an even thinner shadow set. We state the finite and ergodic forms of the remaining problem and locate it within the $\times 2 / \times 3$ rigidity landscape. The conjecture remains open.

1 Introduction

The Collatz conjecture, in Terras form, concerns $T(n) = n/2$ (n even), $T(n) = (3n + 1)/2$ (n odd), and asserts that every positive integer reaches 1. Its two failure modes are nontrivial cycles — excluded for all lengths up to 183 halvings, conditionally on the verified range, in the companion paper [1] — and *divergent orbits*, which are the subject here.

The companion paper establishes a corridor for divergence: an orbit segment avoiding descent must run odd-step density above the critical value $\log 2 / \log 3 = 0.6309\dots$ while within the $\log_2 n$ -bit information budget of its start; densities just above critical are achieved by genuine integers at every scale; and no certificate carrying a bounded amount of information about n can decide who survives. The present paper asks what survival costs on the *other*

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side of the arithmetic — in base 3 — and finds that the cost is precise, quantitative, and of a recognizable kind: a divergent orbit must *compress its own 3-adic digit stream*, forever, by a fixed positive margin.

The mechanism is an exact identity. Writing $j_T(n)$ for the number of odd steps among the first T and abbreviating $j = j_T(n)$,

$$2^T T^T(n) = 3^j n + d_T(n), \quad (1)$$

where the correction $d_T(n)$ — an explicit sum of products of powers of 2 and 3 along the orbit — depends only on the *parity word* w of n (the odd/even pattern of its first T steps), not on n itself. Reducing (1) modulo 3^j annihilates the term $3^j n$:

$$T^T(n) \equiv 2^{-T} d(w) \pmod{3^j}.$$

The 3-adic residue of the endpoint is a function of the word alone. We call it the *shadow* of the window.

Counting shadows against words yields the reduction. By the corridor, a surviving window forces a supercritical word, and supercritical words of length T number at most $\sum_{j \geq 0.631T} \binom{T}{j} \approx 2^{H(0.6309)T} = 2^{0.950T}$. But the shadow lives modulo 3^j , and — by an identity that strikes us as the heart of the matter —

$$\delta \cdot \log_2 3 = 1 \quad \text{exactly at} \quad \delta = \frac{\log 2}{\log 3} :$$

at the critical density the orbit *mints exactly one bit of 3-adic digit information per step*, while its word supplies at most 0.95 bits per step. A divergent orbit must therefore emit 3-adic digits carrying at least 5% less information than their length — sustained compression — and this is forced at every scale simultaneously.

Contributions

1. **The shadow theorems** (Section 3), machine-verified: the exact form (1) with the recursion for d ; word-determinacy of d ; the shadow congruence; and the cocycle law

$$d(w_1 w_2) = 3^{j_2} d(w_1) + 2^{|w_1|} d(w_2),$$

which is the $\mathbb{Z}_2 \times \mathbb{Z}_3$ skew product written in integers and governs how shadows couple across scales.

2. **The density floor** (Section 4), machine-verified and to our knowledge not in the formal literature: if all values of an orbit stay $\geq N$ and the window does not descend, then $2^T N^j \leq (3N + 1)^j$ — odd density at least $\log 2 / \log \frac{3N+1}{N}$, tending to critical as $N \rightarrow \infty$. This is the finite form of “divergent orbits have $\liminf$ parity density $\geq \log 2 / \log 3$,” a statement deliberately *not* delivered by the worst-case master inequalities of [1].
3. **The confinement count** (Section 5), machine-verified: the shadows of all supercritical windows of length T form an explicit set of at most $3^T / 2^{\lfloor 17(T-1)/27 \rfloor}$ elements, while every member’s fiber holds more than 2^{T-1} residues. The constants $17/27$ and 3^T are *identical* to those in the 2-adic Terras theorem of [1]: the two mirrors are one phenomenon.

4. **The reduction** (Section 6): finite and ergodic statements of “divergence = sustained compression,” locating the remaining problem in the joint $\times 2/\times 3$ rigidity landscape, with an explicit numeric gap.
5. **Experiments** (Section 7): the true shadow entropy undercuts the binomial bound (the word-to-shadow map has persistent collisions); the full-window shadow nearly determines the half-window shadow (1.09 prefixes per full value at $2T = 18$) — cross-scale rigidity, as the cocycle predicts; and Sturmian words at near-critical slope occupy an even thinner shadow set (167 distinct shadows from 400 words, against 400 of 400 for unstructured words of matched density).

Everything in Sections 3–5 is formalized in Lean 4 over `mathlib` [4] with no `sorry` and no axioms beyond standard foundations (Section 8).

2 Preliminaries

Notation follows [1]. For $T, n \in \mathbb{N}$ the parity word of n at length T is $w = (T^t(n) \bmod 2)_{t < T} \in \{0, 1\}^T$, and $j_T(n) = \sum_t w_t$. Two facts from [1] are used throughout: the *parity bijection* (the word of length T is determined by, and determines, $n \bmod 2^T$), and the *binomial law* (exactly $\binom{T}{j}$ residues mod 2^T have j odd steps). We also use the dichotomy of [1]: if $2^T \leq n$ and $n \leq T^t(n)$ for all $t \leq T$, then $2^T < 2 \cdot 3^{j_T(n)}$ (“survivors inside the information budget run supercritical”).

3 The shadow

Definition 3.1 (`dcoef`). Set $d_0(n) = 0$ and

$$d_{T+1}(n) = \begin{cases} 3^{j_T(n)} + 2 d_T(n) & n \text{ odd}, \\ 2 d_T(n) & n \text{ even}. \end{cases}$$

Theorem 3.2 (`terras_exact_form`). $2^T T^T(n) = 3^{j_T(n)} n + d_T(n)$ for all T, n .

Proof. Induction on T via the exact doubling identities $2 T(n) = n$ (n even) and $2 T(n) = 3n+1$ (n odd). $\square$

The master inequalities of [1] are the two-sided bounds $3^j - 2^j \leq d_T(n) + (3^j - 2^T) \cdot 0 \leq \dots$ — more usefully: Theorem 3.2 is the identity those inequalities sandwich, and it isolates all n -independence in d :

Theorem 3.3 (`dcoef_modEq`). If $n \equiv m \pmod{2^T}$ then $d_T(n) = d_T(m)$: the correction is a function of the parity word.

Proof. Induction on T ; congruence mod 2^{T+1} propagates through one Terras step to congruence mod 2^T (doubling identities again), and equal parities make the two branches of the recursion agree, using also the word-determinacy of $j_T(\cdot)$ from the parity bijection. $\square$

Theorem 3.4 (the shadow congruence; `shadow_modEq`). *If $n \equiv m \pmod{2^T}$ then*

$$T^T(n) \equiv T^T(m) \pmod{3^{j_T(n)}}.$$

Equivalently: $T^T(n) \bmod 3^j$ depends only on $n \bmod 2^T$, namely $T^T(n) \equiv 2^{-T} d(w) \pmod{3^j}$.

Proof. Two instances of Theorem 3.2 share j and d (Theorem 3.3); adding $3^j m$ to one and $3^j n$ to the other gives $2^T T^T(n) + 3^j m = 2^T T^T(m) + 3^j n$, whence $2^T T^T(n) \equiv 2^T T^T(m) \pmod{3^j}$, and 2^T is invertible modulo 3^j . $\square$

Theorem 3.5 (the cocycle law; `dcoef_add`). *For all S, T, n , writing $n' = T^S(n)$ and $j_2 = j_T(n')$,*

$$d_{S+T}(n) = 3^{j_2} d_S(n) + 2^S d_T(n').$$

Proof. Expand $2^{S+T} T^{S+T}(n)$ through the two windows by Theorem 3.2 and cancel, using additivity of $j()$ over concatenated windows. $\square$

Remark 3.6. *Theorem 3.5 is the $\mathbb{Z}_2 \times \mathbb{Z}_3$ skew product in integer clothing: the second window's 3-adic weight 3^{j_2} scales the first correction (a 3-adic shift), the first window's 2-adic length 2^S scales the second (a 2-adic shift). It also explains why shadows at different scales are strongly coupled (Section 7): modulo $3^{j_1+j_2}$, the full-window shadow contains the first-window correction with multiplier 3^{j_2} .*

4 The density floor

The dichotomy of [1] pins survivors to supercritical density only while inside the $\log_2 n$ -bit budget; beyond the budget, the worst-case inequalities go silent — necessarily so, by the no-go theorem. For *divergent* orbits a different, average-case mechanism takes over: once all values are large, the odd step multiplies by at most $\frac{3N+1}{2N} \approx \frac{3}{2}$, and the accounting closes again.

Theorem 4.1 (`orbit_upper_refined`, `density_floor`). *Let $N \geq 1$ and suppose $T^t(n) \geq N$ for all $t < T$. Then with $j = j_T(n)$,*

$$T^T(n) \cdot 2^T N^j \leq n(3N+1)^j;$$

in particular, if also $n \leq T^T(n)$ (no descent), then

$$2^T N^j \leq (3N+1)^j, \quad \text{i.e.} \quad \frac{j}{T} \geq \frac{\log 2}{\log \frac{3N+1}{N}} \xrightarrow{N \rightarrow \infty} \frac{\log 2}{\log 3}.$$

Proof. Induction; the odd step costs exactly the inequality $N(3n+1) \leq n(3N+1)$, which is $N \leq n$. $\square$

Corollary 4.2. *A divergent orbit (one with $T^t(n) \rightarrow \infty$, hence never returning below any fixed N after some time, and in particular never descending below suitable re-anchored starting points) has $\liminf$ parity density $\geq \log 2 / \log 3$. At that density the orbit performs $\delta \log_2 3 \geq 1$ bit of 3-adic digit production per step, with equality exactly on the critical line.*

5 The confinement count

Definition 5.1 (`shadowPair`, `shadowSet`). For a window length T let $\sigma_T(n) = (j_T(n), T^T(n) \bmod 3^{j_T(n)})$, and let Σ_T be the set of values $\sigma_T(r)$ over the supercritical residues $r < 2^T$ (those with $2^T < 2 \cdot 3^{j_T(r)}$).

By Theorem 3.4, $\sigma_T(n) = \sigma_T(n \bmod 2^T)$ for every n .

Theorem 5.2 (`surviving_shadow_mem`, `shadow_compression`). If $2^T \leq n$ and $n \leq T^t(n)$ for all $t \leq T$, then $\sigma_T(n) \in \Sigma_T$ and $2 \cdot 3^{j_T(n)} > 2^T$. Moreover

$$2^{\lfloor 17(T-1)/27 \rfloor} \cdot |\Sigma_T| \leq 3^T.$$

Thus the shadow of every survivor lies in a set of at most $3^T/2^{\lfloor 17(T-1)/27 \rfloor} \approx 2^{0.9554T}$ elements, while its own fiber $\mathbb{Z}/3^j$ offers more than 2^{T-1} residues: an occupancy ratio $\approx 2^{-0.045T}$. The proof rides the binomial law, the image bound $|\Sigma_T| \leq \#\{\text{supercritical residues}\}$, and the integer-only tail estimate $2^m \sum_{j \geq m} \binom{T}{j} \leq 3^T$ of [1] — the *same* estimate, with the *same* threshold $17/27$ from $3^{17} \leq 2^{27}$, that bounds the 2-adic count of non-descending integers. Terras’ theorem and Theorem 5.2 are mirror images: one counts who can survive, the other counts where survivors may stand in base 3.

6 The reduction

Finite form

Collatz divergence requires an integer n whose orbit, for every window length T along a sequence of re-anchored starting points, (i) runs parity density $\geq \log 2 / \log 3 - o(1)$ (Theorem 4.1), hence mints 3-adic residue information at $\geq 1 - o(1)$ bits per step; while (ii) the minted residues carry at most $\log_2 3 - \frac{17}{27} = 0.9554 \dots$ bits per step of information (Theorem 5.2), since they are functions of a supercritical word. *Divergence is a sustained $\approx 4.5\%$ compression of the orbit’s own 3-adic digit stream* (5.0% at the entropy-optimal constants), enforced consistently across all scales by the cocycle law.

Ergodic form

Embed the orbit diagonally in $\mathbb{Z}_2 \times \mathbb{Z}_3$ and pass to weak-* limits of empirical measures. The limit is invariant for the skew product in which the base ($\mathbb{Z}_2$, conjugate to the parity shift by the bijection of [1]) drives the fiber $\mathbb{Z}_3$ by maps that are 3-adic isometries on even steps and contractions by $|3|_3 = \frac{1}{3}$ on odd steps. For a divergent orbit the base entropy is at most $H(\delta) \leq H(0.6309) = 0.950$ bits per step while the fiber contracts at $\delta \log_2 3 \geq 1$ bit per step, so the limit measure’s $\mathbb{Z}_3$ -marginal has (upper) dimension at most $H(\delta)/(\delta \log_2 3) \leq 0.95 < 1$.

Question 6.1 (the missing rigidity). *Must the $\mathbb{Z}_3$ -marginal of any weak-* limit of empirical measures of a divergent integer orbit have full dimension?*

An affirmative answer excludes divergent orbits. The question sits squarely in the joint $\times 2 / \times 3$ tradition of Furstenberg [2] and Rudolph–Johnson [5]: the integers’ base-2 and base-3 structures are expected to be “independent,” and Question 6.1 asks for one quantitative

consequence. We emphasize the calibration: nothing weaker than dimension > 0.9554 is needed, and the bound 0.95 from the dynamics leaves a genuine, fixed gap. We also emphasize honesty about difficulty: even the much weaker assertion that the binary digits of 3^k are not eventually structured (e.g. popcount growth) is open [6, 7]; Question 6.1 is of that family. By the no-go theorem of [1], no bounded-information argument resolves it; by `parity_pattern_realized`, the compressed shadow sets are inhabited at every finite scale, so no counting argument alone can either.

7 Experiments

Three measurements sharpen the picture beyond the theorems (script `analysis/shadow_experiments.py`; deterministic seeds).

True shadow entropy

Restricting to words of length T with exactly $j = \lceil T \log 2 / \log 3 \rceil$ odd steps:

T	j	words $\binom{T}{j}$	distinct shadows	$\frac{1}{T} \log_2(\text{shadows})$
12	8	495	325	0.695
15	10	3,003	1,915	0.727
18	12	18,564	11,522	0.750
21	14	116,280	70,489	0.767

The word-to-shadow map has persistent collisions (~ 0.035 bits per step below the word-count rate at these lengths): the true compression exceeds the formalized bound. (Caveat: these are small T ; both rates are still rising toward their asymptotes, and we do not claim the gap survives the limit.)

Cross-scale rigidity

At $2T = 18$, $j = 12$: the 18,564 supercritical words yield 11,522 distinct full-window shadows, 226 distinct half-window (prefix) shadows — and only 12,522 distinct *pairs* (prefix shadow, full shadow), i.e. 0.5% of the product bound. The full shadow determines the prefix shadow up to 1.09 candidates on average. As the cocycle law predicts, shadows at nested scales are nearly hierarchical; a divergent orbit must satisfy them all *consistently*, a constraint that counting at a single scale does not see and which we believe is unexplored.

Structured words cluster

Mechanical (Sturmian) words at slopes just above critical — the itineraries forced on slowly divergent orbits by Theorem 4.1 together with the dichotomy — produce 167 distinct shadows from 400 samples at $T = 40$, against 400 of 400 for unstructured words of matched density. Structured itineraries inhabit a far thinner shadow set; arithmetic exclusion of structured divergence should be attacked through the shadow.

8 The formalization

Theorems 3.2–5.2 are formalized in `Shadow.lean` (~430 lines) over the companion development [1] (parity bijection, dichotomy, binomial law, tail estimate). The audit: a clean build with no `sorry`; `#print axioms` reports only Lean’s standard foundations for all seven named theorems, and only `propext` with `Quot.sound` — not even choice — for Theorems 3.2 and 3.5. The numerical claims of Section 7 are independent Python computations; the identity (1) and the word-determinacy of the shadow were additionally cross-checked on 2×10^4 random orbits.

9 Outlook

Three levers, in the order we would pull them. (1) *Cross-scale consistency*: formalize and then exploit the near-hierarchy of shadows — the cocycle law makes “the orbit’s shadows cohere at all scales” a concrete system of congruences whose joint solvability over a single integer has never been studied. (2) *Structured words*: combine the clustering observation with the substitution structure of Sturmian words to attack “no slowly divergent orbits” — the thinness of their shadow sets is exactly what a Mahler-style argument wants. (3) *Question 6.1*: the ergodic statement, for specialists in $\times 2 \times 3$ rigidity, now with explicit constants and a fixed dimension gap. The conjecture remains open; what this paper contributes is a precise statement of *what kind* of miracle a counterexample would be: an integer that out-compresses the entropy of its own arithmetic, at every scale, forever.

Acknowledgments

The formalization, the experiments, and this text were produced in collaboration with Claude (Anthropic). The Lean community’s `mathlib` made the formalization routine.

Reproducibility

`lake build` checks every theorem; `#print axioms` reproduces the audit; `analysis/shadow_experiments.py` regenerates Section 7 in minutes on commodity hardware.

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